

Aryan Hemendra Shah  
College of Arts and Sciences, University of Miami  
Miami, FL 33146  
[ahs222@miami.edu](mailto:ahs222@miami.edu)

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The Editors  
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Dear Editors,

I am submitting for your consideration the manuscript “**An open-source GCR dosimetry pipeline for Mars mission risk assessment: organ-specific cancer risk, acute solar particle events, and global uncertainty quantification**” for publication in *Life Sciences in Space Research*.

**What the paper does.** The manuscript presents an open-source Python package (GCRISK) that integrates GCR spectrum generation, CSDA slab transport, organ-specific cancer risk (REID), Solar Energetic Particle acute dosimetry, and global uncertainty quantification via Latin Hypercube Sampling into a single reproducible workflow with 86 automated tests and a public repository.

**What is new.** I am transparent that the individual components — force-field modulation, CSDA transport, REID estimation — are established methods. The contributions are twofold.

First, no currently available open tool combines organ-specific REID, SEP acute risk, and global uncertainty propagation in a single workflow reproducible from a standard `pip install`, with inter-species flux ratios constrained to ACE/CRIS and PAMELA measurements rather than fit freely to a single total-dose calibration point.

Second, the Latin Hypercube Sampling ensemble shows that the radiobiological parameters (quality factor scale, DDREF, and the excess risk/absolute risk scaling factors) together account for  $\sim 88\%$  of REID output variance, while GCR physics parameters contribute  $\sim 12\%$ , quantifying why physics improvements alone cannot close mission risk uncertainty. This is a directly actionable finding for research prioritization.

**Scope and limitations.** The transport model uses CSDA with exponential nuclear attenuation and does not capture heavy-ion fragmentation secondaries. This limitation is stated explicitly and its domain of validity is discussed. The per-species calibration to MSL/RAD is described plainly as calibration, and the validation evidence is presented accordingly. The REID framework follows Cucinotta (2013); the 2023 NASA update is noted as a limitation for future work. The repository also contains a separate LET/RBE proton-therapy benchmarking layer, reported here only as a brief implementation self-consistency check, since it is the

foundation of a distinct follow-up study.

**Relevance to *Life Sciences in Space Research*.** This manuscript is squarely within the journal's scope: human spaceflight radiation risk, Mars mission planning, and space radiation biology.

**Data and code availability.** All source code, input data, and validation scripts are openly available on GitHub at the repository [aryanhshah8/gcrisk](#), and no proprietary datasets are required to reproduce any results.

I confirm that this manuscript has not been submitted elsewhere, that I am the sole author, and that I have no conflicts of interest to declare. I look forward to the reviewers' comments.

Sincerely,

Aryan Hemendra Shah  
College of Arts and Sciences,  
University of Miami  
[ahs222@miami.edu](mailto:ahs222@miami.edu)  
ORCID: 0009-0008-0518-428X